

comprehensive_admin_topology_test

tests/regression/ comprehensive_admin_topology_test .sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T05:00:00Z **Status:**
Active standing regression guard (§11.4.135) for BUGFIX-ADMIN-TOPOLOGY.

Companion (§11.4.18) to the in-source documentation block at the top of the script. Sibling of docs/scripts/port_topology_aware_test.md.

Overview

A standing regression guard that proves tests/comprehensive-test.sh never reports a **PASS** for a host port that is held by a **non-project process**.

Pre-fix, test_ports() and test_admin() decided admin health purely on "is *something* listening on / answering 200 at :58080?". In the CI/host topology the project's proxy-admin container is **unpublished** (internal port only) while :58080 is held by an unrelated whoami service that answers HTTP 200 to *any* path — and even echoes Hostname: proxy-admin. So three checks (Admin port listening, Admin health endpoint, Admin main page) were **false PASSes hitting the foreign service** — a §11.4.68/§11.4.69 fail-open-to-whatever-answers bluff.

The fix gates the verdict on **project-container ownership**: a listening port is proof only if the named project container is running **and** publishes it (podman port <owner> | grep :<port>). A port that is listening but not project-published is a SKIP, never a PASS.

It does **not** grep the source. It extracts the **real** `_port_topology_check()` from `tests/comprehensive-test.sh` and drives it with stubbed `container/runtime/ss`, so it is deterministic and never touches the data plane.

Prerequisites

- `bash` (the extracted `_port_topology_check` uses `[[ ]]`/local; the probe runs under `bash`).
- `awk`, `mktemp`. No network, no container access, no privileges.

Usage examples

```
# GREEN guard (default) – assert the real check refuses the fail-  
open + PASSes owned:
```

```
tests/regression/comprehensive_admin_topology_test.sh  
# exit 0 = PASS
```

```
# RED reproduce – replicate the pre-fix listening=>PASS logic and  
assert the bluff:
```

```
RED_MODE=1 tests/regression/comprehensive_admin_topology_test.sh  
# exit 0 = defect reproduced
```

```
# Runs automatically inside the suite:
```

```
bash tests/run-tests.sh  
# test_regression_guards()
```

§11.4.115 RED_MODE polarity

RED_MODE	What it runs	PASS means
0 (default)	the REAL <code>_port_topology_check()</code> from <code>tests/ comprehensive-test.sh</code>	a foreign-held port (owner not publishing, a non- project process listening) → SKIP, and an owner- published+listening port → PASS → the fix is present (GREEN guard)
1	the PRE-FIX replica (listening => PASS)	the foreign-held port is classified PASS → the fail- open bluff reproduces

A `RED_MODE=1` run that *cannot* reproduce (replica does not PASS the foreign-held port) is a finding per §11.4.7, not a pass.

The honest truth-table the GREEN guard asserts (owner = the named project container, published = `podman port <owner>` shows the port):

published	listening	verdict	meaning
yes	yes	PASS	project service up and serving on the published port
yes	no	FAIL	owner publishes the port but nothing is bound
no	yes	SKIP	a non-project process holds the port — not the project's service (§11.4.68)
no	no	SKIP	service not deployed in this topology (§11.4.3)

Edge cases

- **Fix reverted** (e.g. `_port_topology_check` flipped back to `ss | grep => PASS`) → the GREEN guard reports REGRESSION: ... fail-open bluff reintroduced ... and exits 1. This is the §1.1 paired-mutation behaviour.
- **Admin genuinely published** (proxy-admin publishes :58080) → the guard's owned-scenario cell still asserts PASS, so the fix does not over-SKIP a real admin.
- **Runtime is docker** → the extracted function's docker port branch is exercised identically (the guard stubs `get_runtime`).

Internal behaviour

- `#!/bin/sh, set -eu`; the bash-only extracted function runs inside an explicit bash probe (§11.4.67).
- Extraction: `awk '/^_port_topology_check\(\) \/{f=1} f{print} / ^\}/{if(f){exit}}'` pulls the live function (closing `}` at column 0).
- Writes one evidence file per run under `qa-results/regression/comprehensive_admin_topology/` (gitignored).

Related

- Fix site: `tests/comprehensive-test.sh` — `_port_topology_check()` + `test_ports()` / `test_admin()` (§11.4.68/§11.4.69 ownership gate).
- Sibling guard: `docs/scripts/port_topology_aware_test.md` (same class for `tests/run-tests.sh`).
- `tests/run-tests.sh` — registers this guard via `test_regression_guards()`.
- `docs/issues/fixed/BUGFIXES.md` — BUGFIX-0010.

Last verified

2026-07-01 — sh -n parse-clean; GREEN proves the fix
(FOREIGN_VERDICT=SKIP + OWNED_VERDICT=PASS); RED_MODE=1 reproduces
(FOREIGN_VERDICT=PASS); registered in run-tests.sh (41 pass / 0 fail /
6 skip, exit 0); comprehensive-test.sh full run 35 pass / 0 fail / 8
skip.